

SABIN PANT

Software Developer · System Design First · Full-Stack & Backend Architecture
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PROFESSIONAL SUMMARY

Passionate backend developer focused on system design and reliability. Skilled in defining actors and flows, designing schemas with proper indexing, and addressing concurrency and data integrity challenges before they reach production. Hands-on internship experience, five AWS certifications, and practical expertise in full-stack development using Node.js, NestJS, and PostgreSQL with Prisma/TypeORM. Dedicated to building enterprise-level infrastructure banking systems, payment gateways, and financial data pipelines where correctness, reliability, and scale are essential.

EDUCATION

BSc (Hons) Computing

2024 – Present

Islington College · London Metropolitan University, Kathmandu

WORK EXPERIENCE

Full-Stack Developer Intern

March 2026 – Present

Leaflet Digital Solutions, Kathmandu

- Building data-intensive systems with Node.js, Express, Next.js and PostgreSQL responsible for both architecture decisions and implementation.
- Designing and securing API pipelines with JWT authentication and role-based access control across multitenant applications.
- Producing API documentation with Swagger, enabling clean handoffs and reducing integration time for frontend teams.

PROJECTS

MediLife — Hospital Management System

Java 21 · Jakarta EE · Apache Tomcat 10 · MySQL · Maven · JSP/JSTL · jBCrypt · Gson

- Built a full-featured hospital management platform under strict MVC architecture covering patient registration, doctor approval workflows, appointment scheduling, billing, and medical records across three roles (Admin, Doctor, Patient).
- Engineered a dual booking flow: patient-initiated Quick Consult (direct booking) and Admin-assigned Request Appointment with rejection handling and reassignment logic modelling real-world hospital triage workflows.

- ▶ Implemented secure BCrypt password hashing, session-based authentication with configurable timeout, admin approval workflow for doctor registrations with licence document review, and account lock/unlock with reason tracking.
- ▶ Designed a notification system triggered by booking, confirmation, completion, billing, and account events plus financial reporting with CSV export, system logs, and broadcast announcements for admin oversight.

Nebula Chat — Real-Time Chat Application

TypeScript · Node.js · Express · Socket.IO · TypeORM · PostgreSQL · Redis · React · Tailwind CSS · Docker

- ▶ Built a production-grade real-time chat platform with one-to-one and group messaging, supporting dual authentication flows (OTP-based passwordless login and Google OAuth 2.0 with audience validation against cross-app token spoofing).
- ▶ Engineered real-time message delivery using [Socket.IO](#) with JWT-authenticated WebSocket connections, typing indicators, read receipts, and unread message counts all managed through a singleton connection manager supporting multi-device sessions.
- ▶ Designed a 3NF-normalized PostgreSQL schema (8 tables) with TypeORM, implementing soft-delete participant tracking, self-referencing message replies, and composite primary keys on junction tables for data integrity.
- ▶ Applied strict clean architecture principles controllers handle routing only, services contain all business logic, repositories manage data access with a global exception filter catching all errors and Zod validation at every API boundary.
- ▶ Deployed with Docker Compose (PostgreSQL, Redis, MailHog) using environment-variable-driven configuration with zero hardcoded values and implemented JWT token rotation with httpOnly refresh cookies and in-memory access tokens.

TECHNICAL SKILLS

Backend	Node.js, NestJS, Express.js, Java (JSP, Servlets, DAO, Filters, MVC), TypeScript, Python
Frontend	React, Next.js, Tailwind CSS, HTML5, CSS3, Bootstrap, JSP/JSTL
Databases	PostgreSQL, MySQL, MongoDB, Oracle, Prisma ORM, Redis
Architecture	System design, REST API design, JWT auth, role-based access control, event-driven patterns, BullMQ
Cloud & Infra	AWS (EC2, S3, Lambda), Docker, Vercel, Render, Cloudinary
Tools	Git, GitHub, Postman, Swagger, Figma, PowerBI, Maven
Soft Skills	System design before code, edge case analysis, documentation, fast learner, extreme attention to detail

CERTIFICATIONS

- ▶ **AWS Academy Graduate** – Cloud Foundations, Data Engineering, Machine Learning Foundations, Machine Learning for NLP, Generative AI Foundations
- ▶ **UI/UX with Figma**